

10.7.95

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

Find all circles that touch the x -axis at a distance 3 from the origin and have a y -axis intercept of length $2\sqrt{7}$.

Solution:

$$\mathbf{x}^T \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0, \quad \mathbf{C} = -\mathbf{u}, \quad r^2 = \mathbf{C}^T \mathbf{C} - f \quad (1)$$

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (2)$$

$$\mathbf{C} = \begin{pmatrix} \mathbf{e}_1^T \mathbf{C} \\ \mathbf{e}_2^T \mathbf{C} \end{pmatrix} \quad (3)$$

Substituting y -axis in circle equation

$$(\mathbf{y}\mathbf{e}_2)^T (\mathbf{y}\mathbf{e}_2) + 2\mathbf{u}^T (\mathbf{y}\mathbf{e}_2) + f = 0 \quad (4)$$

$$\implies y^2 - 2(\mathbf{C}^T \mathbf{e}_2)y + f = 0 \quad (5)$$

Here the square of Intercept equals to determinant of (5)

$$\Delta = [-2(\mathbf{C}^T \mathbf{e}_2)]^2 - 4f = 4[(\mathbf{C}^T \mathbf{e}_2)^2 - f] = (2\sqrt{7})^2 \quad (6)$$

$$(\mathbf{C}^T \mathbf{e}_2)^2 - f = 7 \quad (7)$$

$$\mathbf{C}^T \mathbf{C} - f = r^2 \quad (8)$$

$$(8) - (7) \implies \mathbf{C}^T \mathbf{C} - (\mathbf{C}^T \mathbf{e}_2)^2 = r^2 - 7 \implies (\mathbf{e}_1^T \mathbf{C})^2 = r^2 - 7 \quad (9)$$

Since Circle touches x -axis at a distance 3 from origin.

$$|\mathbf{e}_2^T \mathbf{C}| = r \quad (10)$$

$$|\mathbf{e}_1^T \mathbf{C}| = 3 \implies r^2 = 16 \quad (11)$$

$$(\mathbf{e}_2^T \mathbf{C})^2 - f = 7, \implies f = 9 \quad (12)$$

$$\mathbf{C} = \begin{pmatrix} \pm 3 \\ \pm 4 \end{pmatrix} \quad (13)$$

$$\mathbf{x}^\top \mathbf{x} + 2(-\mathbf{C})^\top \mathbf{x} + f = 0 \quad (14)$$

$$\mathbf{x}^\top \mathbf{x} + 2 \begin{pmatrix} -3 & -4 \end{pmatrix} \mathbf{x} + 9 = 0, \quad \mathbf{x}^\top \mathbf{x} + 2 \begin{pmatrix} -3 & 4 \end{pmatrix} \mathbf{x} + 9 = 0, \quad (15)$$

$$\mathbf{x}^\top \mathbf{x} + 2 \begin{pmatrix} 3 & -4 \end{pmatrix} \mathbf{x} + 9 = 0, \quad \mathbf{x}^\top \mathbf{x} + 2 \begin{pmatrix} 3 & 4 \end{pmatrix} \mathbf{x} + 9 = 0 \quad (16)$$

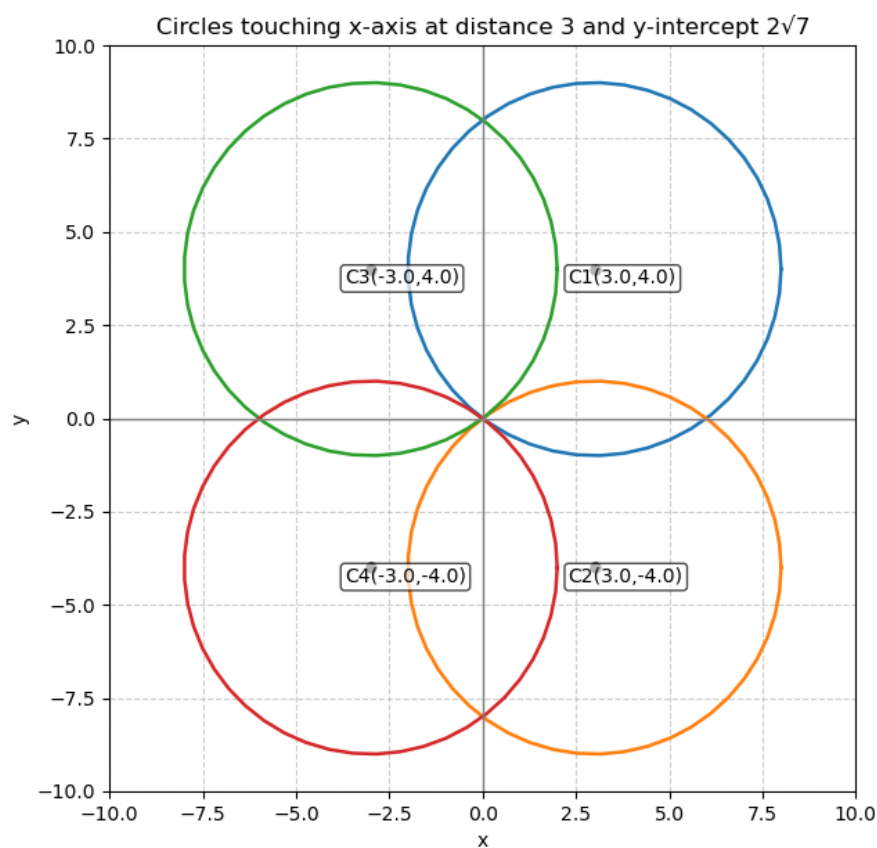


Figure 1